So fraction increase

$$= \frac{B_2 - B_1}{B_1} = \frac{\mu_0(1+\lambda)ni - \mu_0 ni}{\mu_0 ni}$$

$$= \lambda = 1.2 \times 10^{-5}$$

Hence option 1 correct.

Q. 13 Two parallel, long wires are kept 0.20 m apart in a vacuum, each carrying a current of x A in the same direction. If the force of attraction per meter of each wire is 2×10^{-6} N, then the value of x is approximately:

Option 1:

1

Option 2:

2.4

Option 3:

1.4

Option 4:

2

Correct Answer:

1.4

Solution:

$$\frac{dF}{dl} = \frac{\mu o \mathrm{I}_1 I_2}{2\pi d}$$

$$= \frac{4\pi \times 10^{-7} \times x \times x}{2\pi \times 0.2} = 2 \times 10^{-6}$$

$$x = \sqrt{2} = 1.4$$

Hence option 3 is correct

Q. 14 A coil is placed in a time-varying magnetic field. If the number of turns in the coil were to be halved and the radius of wire doubled, the electrical power dissipated due to the current induced in the coil would be :

(Assume the coil to be short circuited.)

Option 1:

Halved

Option 2:

Quadrupled

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Option 3:

$$\mathrm{E}_y = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^3 \left(x - 3 \times 10^8 \mathrm{t}\right)\right] \hat{j} \mathrm{Vm}^{-1}$$

$$\mathrm{B}_z = 60 \sin\left[\frac{\pi}{4} \times 10^3 \left(x - 3 \times 10^8 \mathrm{t}\right)\right] \hat{k}\ \mathrm{T}$$

Option 4:

$$\mathrm{E}_y = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^4 \left(x - 4 \times 10^8 \mathrm{t}\right)\right] \hat{j} \mathrm{Vm}^{-1}$$

$$\mathrm{B}_z = 60 \sin\left[\frac{\pi}{4} \times 10^4 \left(x - 4 \times 10^8 \mathrm{t}\right)\right] \hat{k}\ \mathrm{T}$$

Correct Answer:

$$\mathrm{E}_y = 60 \sin\left[\frac{\pi}{4} \times 10^3 \left(x - 3 \times 10^8 \mathrm{t}\right)\right] \hat{j} \mathrm{Vm}^{-1}$$

$$\mathrm{B}_z = 2 \times 10^{-7} \sin\left[\frac{\pi}{4} \times 10^3 \left(x - 3 \times 10^8 \mathrm{t}\right)\right] \hat{k}\ \mathrm{T}$$

Solution:

$\text{speed of wave in vaccum} = 3 \times 10^8\ \text{m/s}$

$E_0 = CB_0$

$$B_0 = \frac{60}{3 \times 10^8} = 2 \times 10^{-7}\ \mathrm{T}$$

Hence option 2 is correct.

Q. 16 In young's double slit experiment performed using monochromatic light of wavelength λ, when a glass plate $(\mu = 1.5)$ of thickness $\mathrm{x}\lambda$ is introduced in the path of the one of the interfering beams, the intensity at the position where the central maximum occurred previously remains unchanged. The value of x will be :

Option 1:

3

Option 2:

2

Option 3:

1.5

Option 4:

0.5

Correct Answer:

2

Solution:

For the intensity to remain the same the position must be of maxima.

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'so path difference must be $n\lambda$

$(1.5-1)x\lambda = n\lambda$

$x = 2n, n = 0, 1, 2 \ldots$

$x = 0, 2, 4, 6 \ldots$

Hence option B is correct.

Q. 17 Let K_1 and K_2 be the maximum kinetic energies of photo-electrons emitted when two monochromatic beams of wavelength λ_1 and λ_2, respectively are incident on a metallic surface.

$\text{If } \lambda_1 = 3\lambda_2 \text{ then :}$

Option 1:

$\mathrm{K}_1 > \frac{\mathrm{K}_2}{3}$

Option 2:

$\mathrm{K}_1 < \frac{\mathrm{K}_2}{3}$

Option 3:

$\mathrm{K}_1 = \frac{\mathrm{K}_2}{3}$

Option 4:

$\mathrm{K}_2 = \frac{\mathrm{K}_1}{3}$

Correct Answer:

$\mathrm{K}_1 < \frac{\mathrm{K}_2}{3}$

Solution:

Using Einstein's photoelectric effect eqn

$E = \phi + k_{\mathrm{Emx}}$

$\frac{hC}{\lambda_1} = \Phi + K_1 - (1)$

$\frac{hC}{\lambda_2} = \phi + K_2 - (2)$

taking the ratio of Eqn (1) with Eqn (2)

And using $\lambda_1 = 3\lambda_2$

$\frac{\lambda_2}{\lambda_1} = \frac{\phi + K_1}{\phi + K_2} = \frac{1}{3}$

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Correct Answer:
(A) is true but **(R)** is false.

Solution:
Assertion is correct but Reason says that cis-polyisoprene molecules consists of various chains held together by strong polar interactions with coiled structure, which is incorrect.

The chains are held together by weak Vander Waal's interactions.

Hence ,**Option (3)** is Correct.

Q. 19 When sugar 'X' is boiled with dilute H_2SO_4 in alcoholic solution,two isomers 'A' and 'B' are formed 'A' on oxidation with HNO_3 yields saccharic acid where as 'B' is laevorotatory, The compound 'X' is:

Option 1:
Maltose

Option 2:
Sucrose

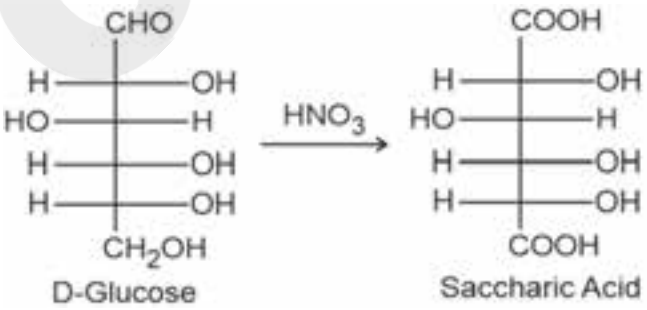
Option 3:
Lactose

Option 4:
Strach

Correct Answer:
Sucrose

Solution:

$$\underset{[\alpha]=66.6^\circ}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\substack{D-Glucose \\ [\alpha]=+52.7^\circ \\ (A)}}{C_6H_{12}O_6} + \underset{\substack{D-Fructose \\ [\alpha]=-92.2^\circ \\ (B)}}{C_6H_{12}O_6}$$

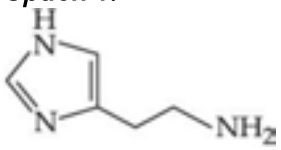


Hence, **Option (2)** is correct.

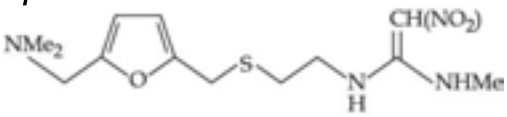
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Q. 20 The drug tegamet is:

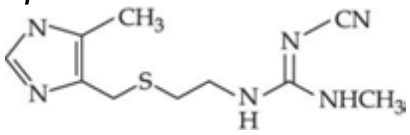
Option 1:



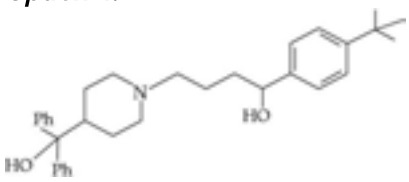
Option 2:



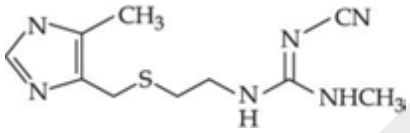
Option 3:



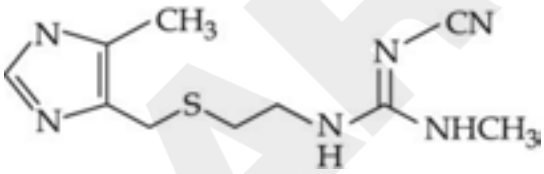
Option 4:



Correct Answer:



Solution:
Tegamet is name of cimetidine which is



Hence **Option (3)** is correct.

Q. 21 100 g of an ideal gas is kept in a cylinder of 416 L volume at $27^{\circ}\mathrm{C}$ under 1.5 bar pressure. The molar mass of the gas is _____ $\mathrm{gmol}-1$ (Nearest integer)

$$(\text{Given : R} = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1})$$

Correct Answer:
4

Solution:

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$\Rightarrow \alpha = 30, \beta = -4$

$\Rightarrow \alpha + \beta = 26$

Hence answer is 26

Q. 22 Let $y = y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + \frac{\sqrt{2}y}{2\cos^4 x - \cos 2x} = xe^{\tan^{-1}(\sqrt{2}\cot 2x)}, 0 < x < \pi/2 \text{ with } y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32}.$$

If $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{18}e^{-\tan^{-1}(\alpha)}$, then the value of $3\alpha^2$ is equal to_____.

Correct Answer:

2

Solution:

$$\frac{dy}{dx} + \frac{\sqrt{2}y}{2\cos^4 x - \cos 2x} = xe^{\tan^{-1}(\sqrt{2}\cot 2x)}$$

Now to get I.F, first find

$$\int \frac{\sqrt{2}}{2\cos^4 x - \cos 2x}dx$$

$$= \int \frac{\sqrt{2}}{2\cos^4 x - 2\cos^2 x + 1}dx$$

$$= \int \frac{\sqrt{2}}{2\cos^2 x(\cos^2 x - 1) + 1}dx$$

$$= \int \frac{\sqrt{2}}{-2\sin^2 x\cos^2 x + 1}dx$$

$$= \int \frac{2\sqrt{2}}{2 - \sin^2(2x)}dx$$

Dividing by $\cos^2 2x$

$$= \int \frac{2\sqrt{2}\sec^2 2x}{2\sec^2 2x - \tan^2(2x)}dx$$

$$= \int \frac{2\sqrt{2}\sec^2 2x}{2 + \tan^2(2x)}dx$$

Let $\tan 2x = t \Rightarrow 2\sec^2 2xdx = dt$

$$= \int \frac{\sqrt{2}dt}{t^2 + 2} = \tan^{-1}\left(\frac{t}{\sqrt{2}}\right)$$

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$$= \tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)$$

$$\therefore \text{ I.F. } = e^{\tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)}$$

Solution of equation is

$$y \cdot e^{\tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)} = \int x e^{\tan^{-1}(\sqrt{2}\cot 2x)} \cdot e^{\tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)} dx$$

$$y \cdot e^{\tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)} = \int x e^{\pi/2} dx$$

$$(\text{As } \sqrt{2}\cot x \cdot \frac{1}{\sqrt{2}}\tan x = 1\ )$$

$$\Rightarrow y = \frac{x^2}{2} e^{\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)} + ce^{-\tan^{-1}\left(\frac{1}{\sqrt{2}}\tan 2x\right)}$$

$$\Rightarrow y = \frac{x^2}{2} e^{\frac{\pi}{2} - \cot^{-1}(\sqrt{2}\cot 2x)} + ce^{-cot^{-1}(\sqrt{2}\cot 2x)}$$

$$\Rightarrow y = \frac{x^2}{2} e^{\tan^{-1}(\sqrt{2}\cot 2x)} + ce^{-\cot^{-1}(\sqrt{2}\cot 2x)}$$

$$\text{As } y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{32} \Rightarrow \frac{\pi^2}{32} = \frac{\pi^2}{32} + c \cdot e^{-\pi/2} \Rightarrow c = 0$$

$$\text{Put } x = \frac{\pi}{3}$$

$$y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{18} e^{\tan^{-1}\left(-\frac{\sqrt{2}}{\sqrt{3}}\right)} \Rightarrow \alpha = \sqrt{\frac{2}{3}} \Rightarrow 3\alpha^2 = 2$$

Hence answer is 2

Q. 23 Let d be the distance between the foot of perpendiculars of the points $P(1, 2, -1)$ and $Q(2, -1, 3)$ on the plane $-x + y + z = 1$. Then d^2 is equal to_____.

Correct Answer:
26

Solution:
$P(1, 2, -1)$

Let $P'(a, b, c)$ be its foot of perpendicular

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